Practical 1 - Predict a Student's Marks from the Number of Hours Studied Using Simple Linear Regression.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, r2_score

df = pd.read_csv("student_marks.csv")

X = df[["StudyHours"]]
y = df["Marks"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
print("Actual Marks:", list(y_test))
print("Predicted Marks:", list(y_pred.round(2)))
print("MAE:", round(mean_absolute_error(y_test, y_pred), 2))
print("R2 Score:", round(r2_score(y_test, y_pred), 2))

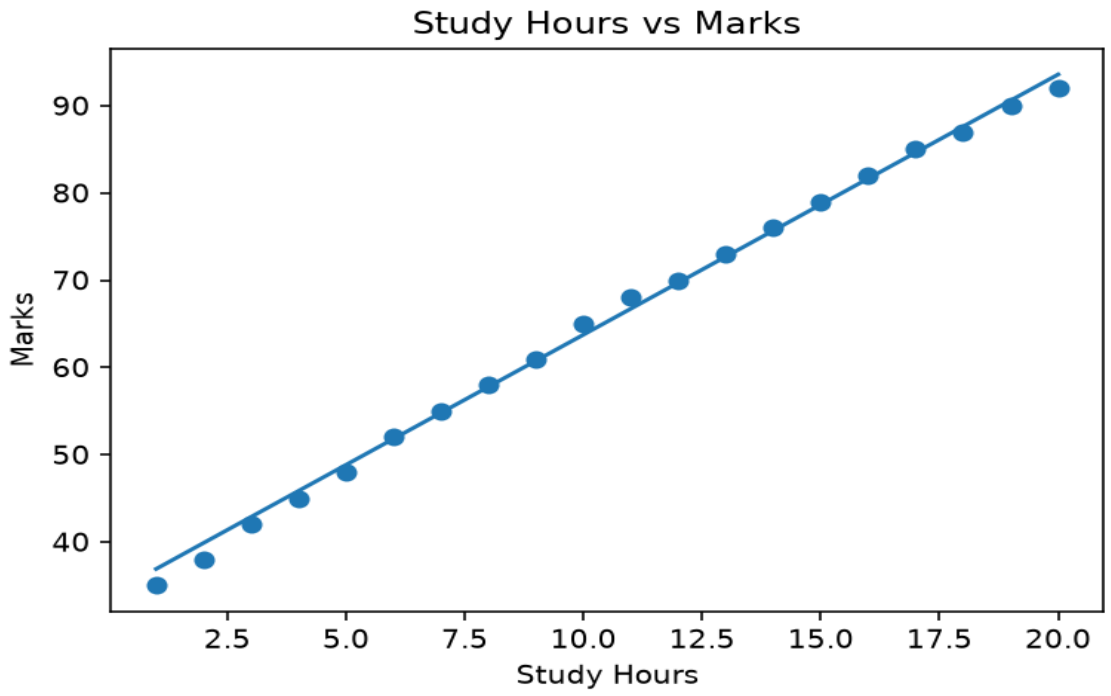
new_student = [[12]]
print("Predicted marks for 12 study hours:",
      round(model.predict(new_student)[0], 2))

plt.scatter(X, y)
plt.plot(X, model.predict(X))
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Study Hours vs Marks")
plt.show()
```

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Output:

```
In [2]: %runfile C:/Users/coolv/OneDrive/Desktop/MLP/untitled0.py --wdir
Slope: 2.9837128638214447
Intercept: 33.92610465993063
Actual Marks: [35, 87, 82, 38]
Predicted Marks: [np.float64(36.91), np.float64(87.63), np.float64(81.67),
np.float64(39.89)]
MAE: 1.19
R2 Score: 1.0
Predicted marks for 12 study hours: 69.73
C:\Users\coolv\miniforge3\envs\cropml\Lib\site-
packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid
feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```



Practical 2. Predict a House's Price from its Area Using Simple Linear Regression.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, r2_score

df = pd.read_csv("house_price.csv")

X = df[["AreaSqFt"]]
y = df["PriceLakh"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

pred = model.predict(X_test)

print("Regression Equation:")
print("Price =", round(model.intercept_, 2),
      "+", round(model.coef_[0], 4), "* Area")

print("Actual Prices:", list(y_test))
print("Predicted Prices:", list(pred.round(2)))
print("MAE:", round(mean_absolute_error(y_test, pred), 2))
print("R2 Score:", round(r2_score(y_test, pred), 2))

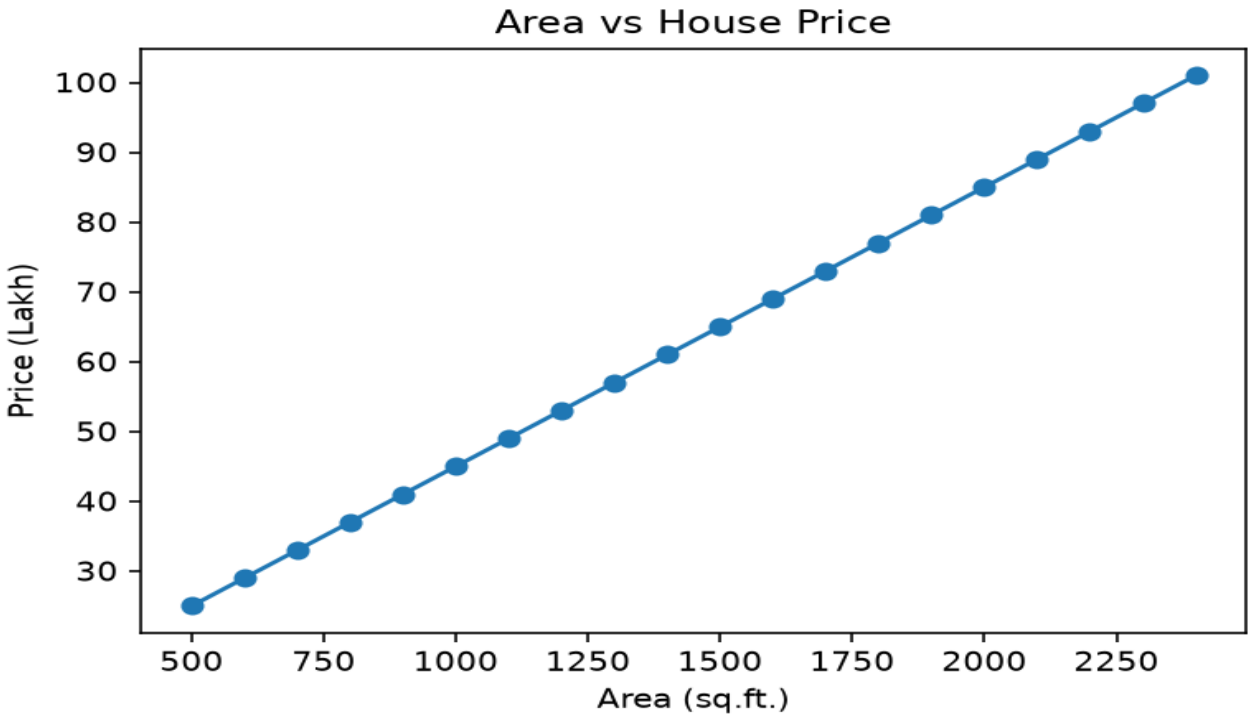
new_house = [[1500]]
print("Predicted price for 1500 sq.ft.:",
      round(model.predict(new_house)[0], 2), "Lakh")

plt.scatter(X, y)
plt.plot(X, model.predict(X))
plt.xlabel("Area (sq.ft.)")
plt.ylabel("Price (Lakh)")
plt.title("Area vs House Price")
plt.show()
```

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Output:

```
In [3]: %runfile C:/Users/coolv/OneDrive/Desktop/MLP/untitled1.py --wdir
Regression Equation:
Price = 5.0 + 0.04 * Area
Actual Prices: [25, 93, 85, 29]
Predicted Prices: [np.float64(25.0), np.float64(93.0), np.float64(85.0),
np.float64(29.0)]
MAE: 0.0
R2 Score: 1.0
Predicted price for 1500 sq.ft.: 65.0 Lakh
C:\Users\coolv\miniforge3\envs\cropml\Lib\site-
packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid
feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```



Practical 3. Predict a Student's Exam Score from Attendance Percentage Using Simple Linear Regression.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

df = pd.read_csv("attendance_score.csv")

X = df[["Attendance"]]
y = df["ExamScore"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

pred = model.predict(X_test)

print("Slope:", round(model.coef_[0], 3))
print("Intercept:", round(model.intercept_, 3))
print("Actual:", list(y_test))
print("Predicted:", list(pred.round(2)))

mse = mean_squared_error(y_test, pred)
print("MSE:", round(mse, 2))
print("R2 Score:", round(r2_score(y_test, pred), 2))

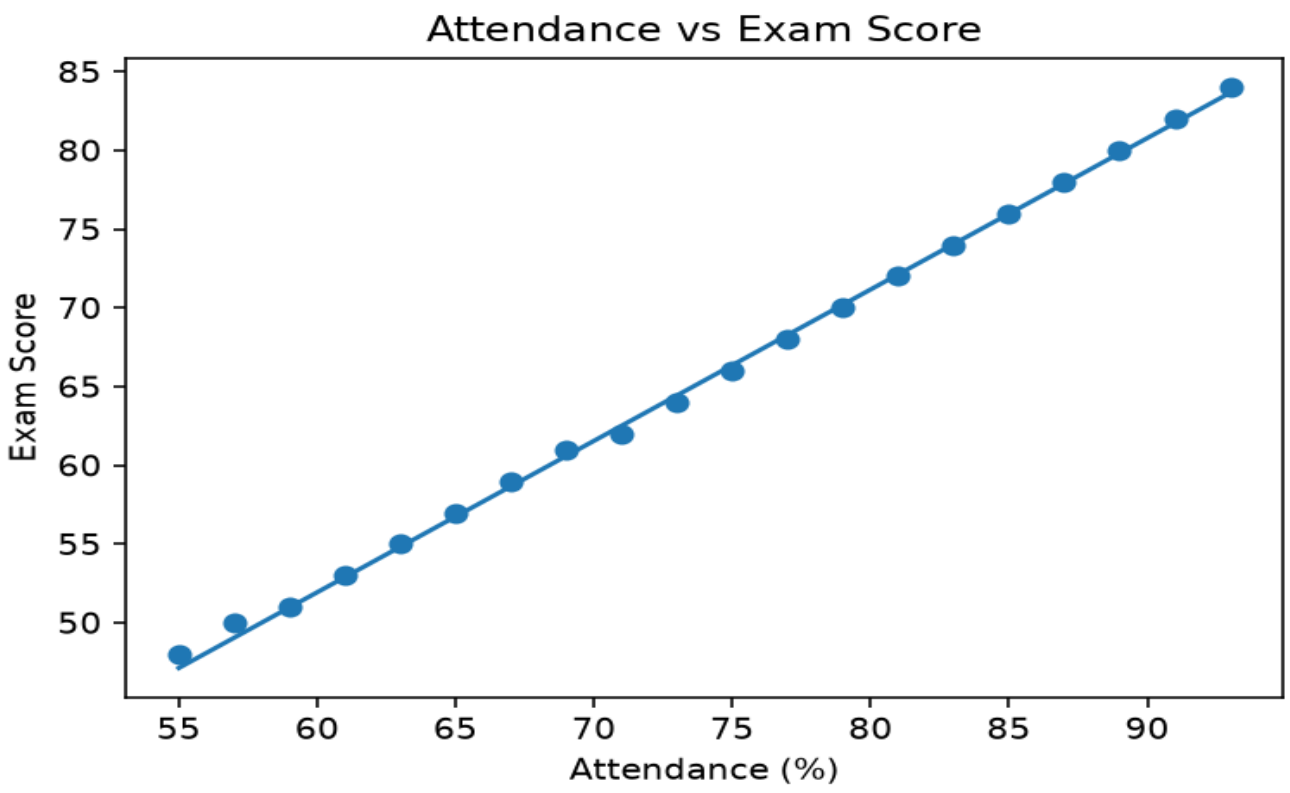
new_student = [[80]]
print("Predicted score for 80% attendance:",
      round(model.predict(new_student)[0], 2))

plt.scatter(X, y)
plt.plot(X, model.predict(X))
plt.xlabel("Attendance (%)")
plt.ylabel("Exam Score")
plt.title("Attendance vs Exam Score")
plt.show()
```

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Output:

```
In [4]: %runfile C:/Users/coolv/OneDrive/Desktop/MLP/untitled2.py --wdir
Slope: 0.962
Intercept: -5.755
Actual: [48, 80, 76, 50]
Predicted: [np.float64(47.13), np.float64(79.82), np.float64(75.98),
np.float64(49.05)]
MSE: 0.42
R2 Score: 1.0
Predicted score for 80% attendance: 71.17
C:\Users\coolv\miniforge3\envs\cropml\Lib\site-
packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid
feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```



Practical 4. Predict Monthly Sales from Advertising Expenditure Using Simple Linear Regression.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, r2_score

df = pd.read_csv("advertising_sales.csv")

X = df[["Advertising"]]
y = df["MonthlySales"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

pred = model.predict(X_test)

print("Slope:", round(model.coef_[0], 3))
print("Intercept:", round(model.intercept_, 3))
print("Actual Sales:", list(y_test))
print("Predicted Sales:", list(pred.round(2)))
print("MAE:", round(mean_absolute_error(y_test, pred), 2))
print("R2 Score:", round(r2_score(y_test, pred), 2))

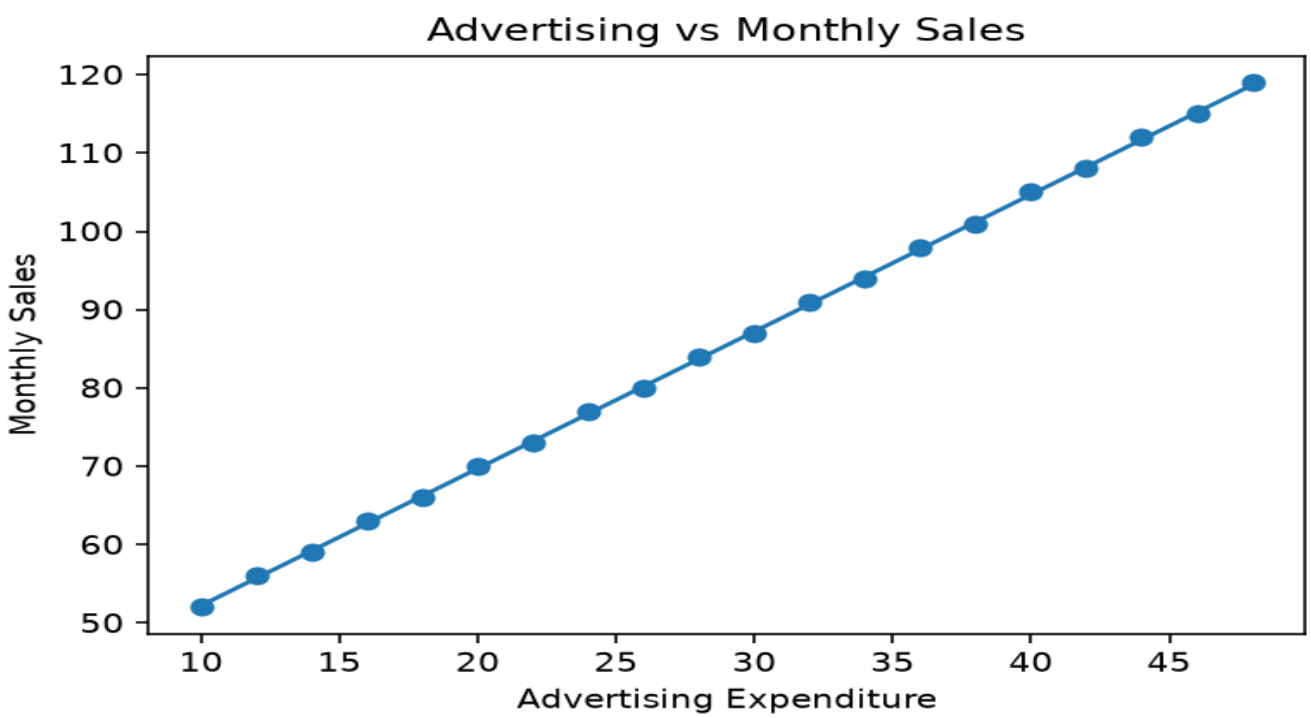
new_advertising = [[35]]
print("Predicted sales for advertising = 35:",
      round(model.predict(new_advertising)[0], 2))

plt.scatter(X, y)
plt.plot(X, model.predict(X))
plt.xlabel("Advertising Expenditure")
plt.ylabel("Monthly Sales")
plt.title("Advertising vs Monthly Sales")
plt.show()
```

Vansh Prajapati

Output:

```
In [5]: %runfile C:/Users/coolv/OneDrive/Desktop/MLP/untitled3.py --wdir
Slope: 1.749
Intercept: 34.749
Actual Sales: [52, 112, 105, 56]
Predicted Sales: [np.float64(52.24), np.float64(111.7), np.float64(104.71),
np.float64(55.74)]
MAE: 0.27
R2 Score: 1.0
Predicted sales for advertising = 35: 95.96
C:\Users\coolv\miniforge3\envs\cropml\Lib\site-
packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid
feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```



Practical 5. Predict Fuel Consumption from Distance Travelled Using Simple Linear Regression.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, r2_score

df = pd.read_csv("fuel_consumption.csv")

X = df[["DistanceKm"]]
y = df["FuelLitres"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

pred = model.predict(X_test)

print("Slope:", round(model.coef_[0], 4))
print("Intercept:", round(model.intercept_, 4))
print("Actual Fuel:", list(y_test))
print("Predicted Fuel:", list(pred.round(2)))
print("MAE:", round(mean_absolute_error(y_test, pred), 2))
print("R2 Score:", round(r2_score(y_test, pred), 2))

new_distance = [[250]]
print("Predicted fuel for 250 km:",
      round(model.predict(new_distance)[0], 2), "litres")

plt.scatter(X, y)
plt.plot(X, model.predict(X))
plt.xlabel("Distance (km)")
plt.ylabel("Fuel Consumption (litres)")
plt.title("Distance vs Fuel Consumption")
plt.show()
```

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Output:

```
In [6]: %runfile C:/Users/coolv/OneDrive/Desktop/MLP/untitled3.py --wdir
Slope: 0.0713
Intercept: 0.3889
Actual Fuel: [1.8, 26.0, 23.2, 3.2]
Predicted Fuel: [np.float64(1.81), np.float64(26.04), np.float64(23.19),
np.float64(3.24)]
MAE: 0.03
R2 Score: 1.0
Predicted fuel for 250 km: 18.21 litres
C:\Users\coolv\miniforge3\envs\cropml\Lib\site-
packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid
feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```

